OpenDRIVE® 1.8.0, Static Road Network Description, 2023-11-22

10 Crossings and cross paths

10.1 Introduction

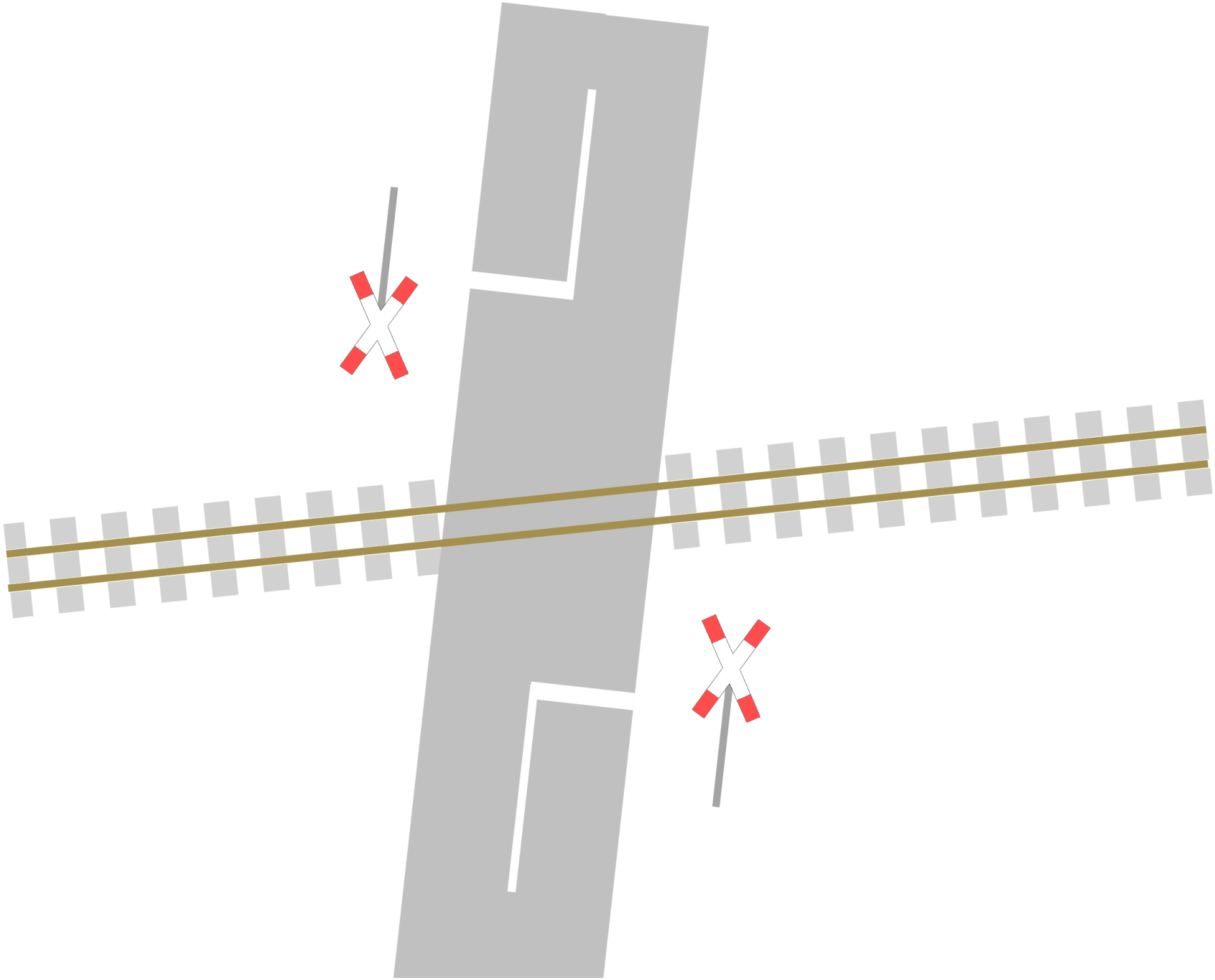
Crossings and cross paths are minimal concepts for junctions where traffic of two roads can cross.

Crossings are junctions without connecting roads and only a definition of the sections where crossing traffic can appear. Traffic cannot change the roads at crossings.

Cross paths are intended for pedestrian crossings and are junction elements where traffic of a lane can cross other lanes and continue on a different lane of the same or a different road.

Typical examples for a crossings or cross paths are:

- Pedestrian crossings
- Bicycle crossings
- Rail road crossings



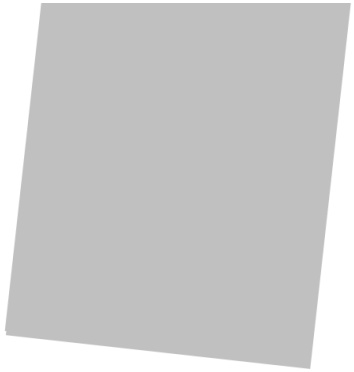


Figure 58. A rail road crossing

Figure 58 shows a rail road crossing.

NOTE

Pedestrian crossings within common junctions are described in Section 6.3.3, "Advanced T junction".

10.2 Rules and guidelines

- Crossings are intended for pedestrians, cyclists, or rail bound vehicles.
- Traffic cannot change the roads at crossings.
- Cross paths are intended for pedestrians and cyclists.

10.3 Pedestrian crossing

10.3.1 Example

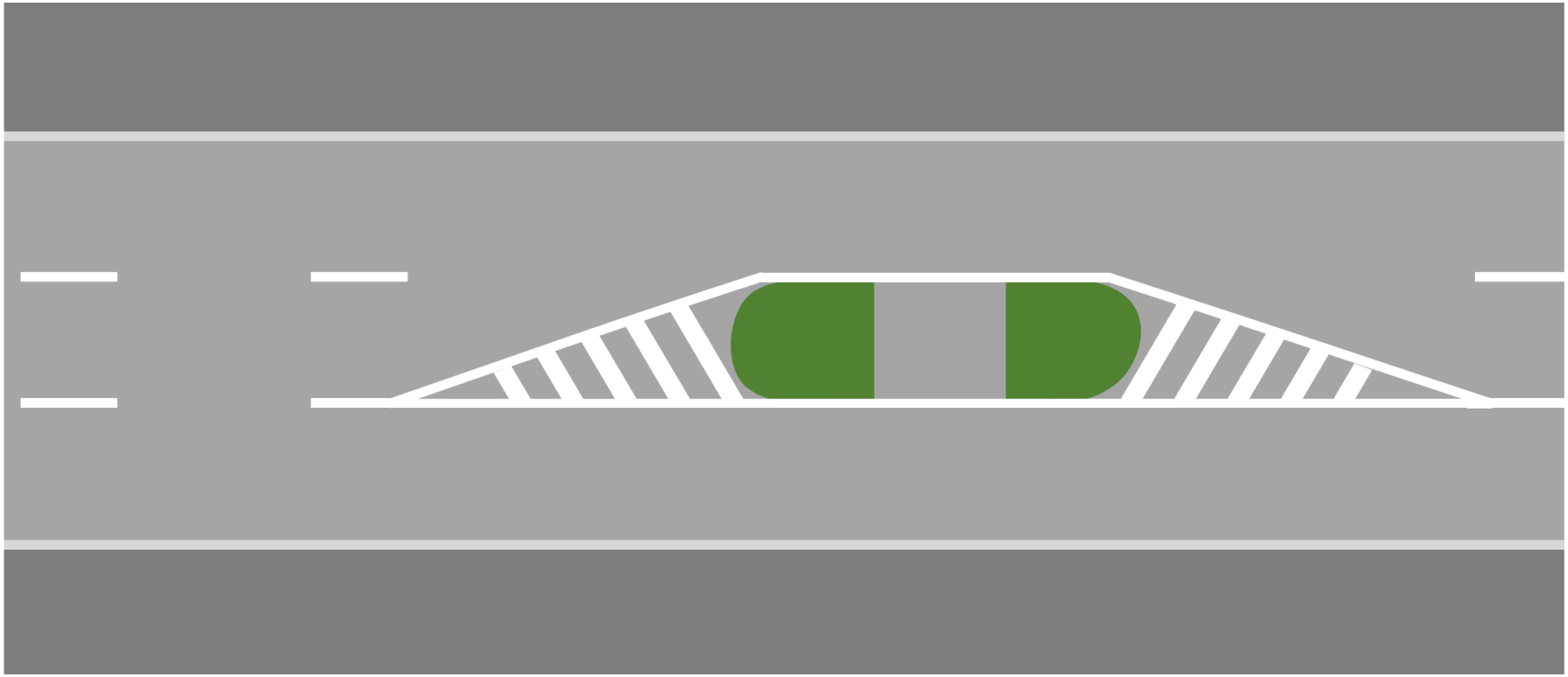


Figure 59. Pedestrian crossing

Figure 59 shows a pedestrian crossing.

10.3.2 Challenges

- Choose between a crossing or a cross path to find the optimal junction type.
- Decide where additional roads are necessary.
- Assign the correct lane types.
- Set the range for the area of crossing traffic.
- Create all necessary lane links.

10.3.3 Modeling approach

A junction with @type="virtual" and <crossPath> elements is the best choice for connections between sidewalks of different roads.

1. Place the road reference line for the road that is for general traffic.

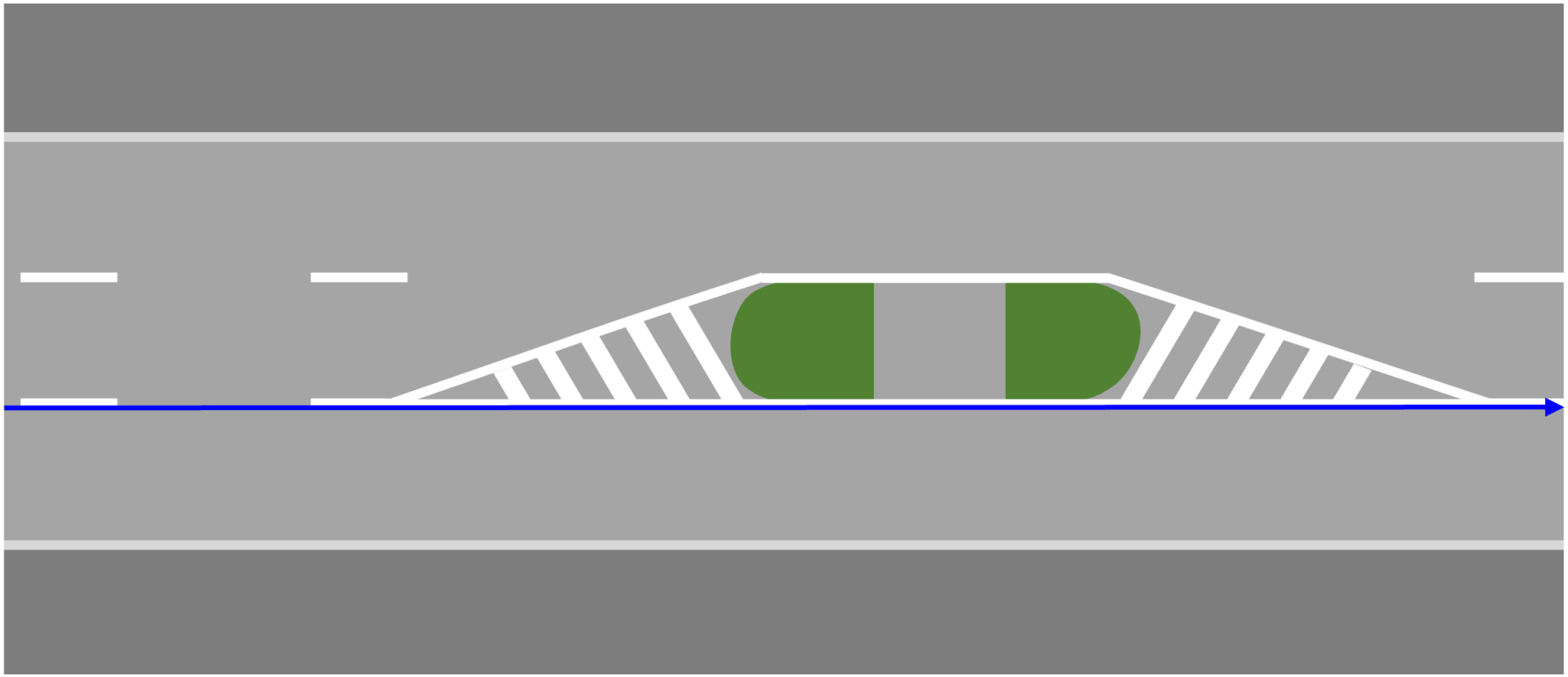


Figure 60. Road reference line

Figure 60 shows the road reference line (blue arrow) of the road for general traffic.

Code 49. Road for general traffic at the pedestrian crossing

```
<road name="drivingRoad" length="21" id="1">
    ...
</road>
```

2. Add one lane with @type="driving" in the direction of the road reference line and two lanes with @type="driving" in the opposite direction, one lane with @type="curb" at the outside of the driving lanes, and next to the curb one lane with @type="walking" on each side of the road.

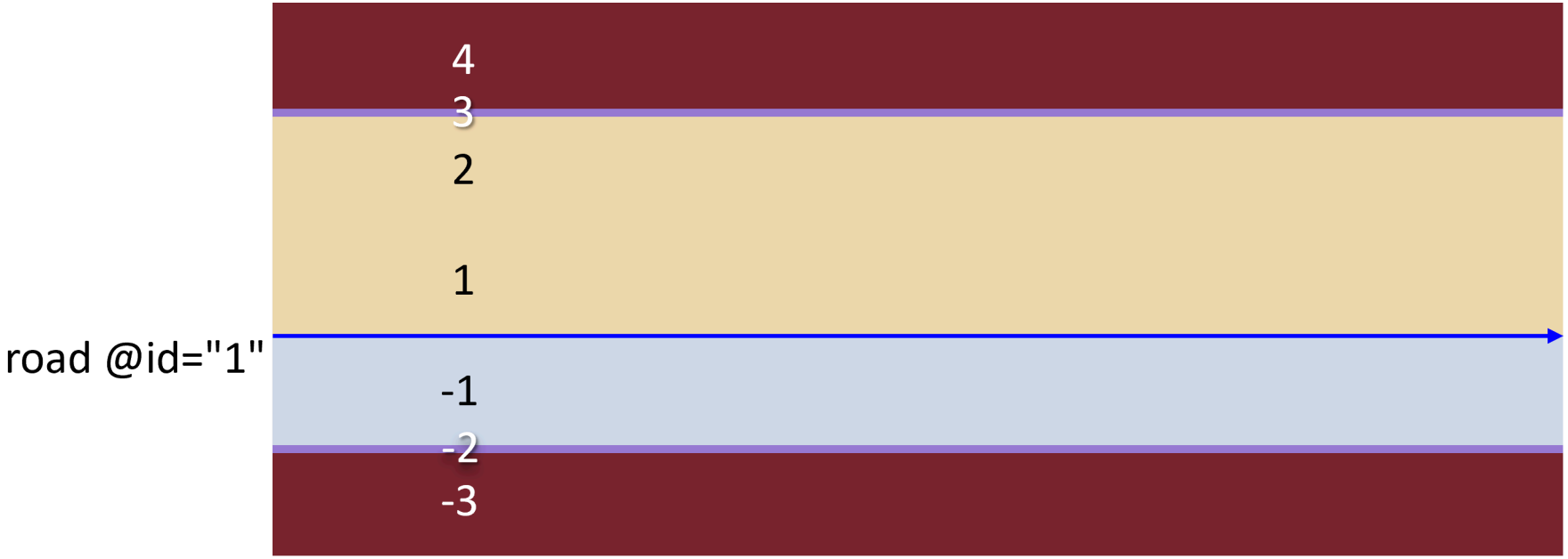


Figure 61. Lanes on the road for general traffic

Figure 61 shows the road for general traffic with the usual lane types.

Code 50. Lanes on the road for general traffic

```xml
<road name="drivingRoad" length="1.2000000000000000e+02" id="1">
    ...
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left>
                <lane id="4" type="walking">
                    ...
                </lane>
                <lane id="3" type="curb">
                    ...
                </lane>
                <lane id="2" type="driving">
                    ...
                </lane>
                <lane id="1" type="driving">
                    ...
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    ...
                </lane>
                <lane id="-2" type="curb">
                    ...
                </lane>
                <lane id="-3" type="walking">
                    ...
                </lane>
            </right>
        </laneSection>
    </lanes>
    ...
</road>
```

```
3
2
4
2
5
2
6
2
7
2
8
2
9
3
0
3
1
3
2
3
3
3
4
```

3. Create two lane sections and add a lane with @type="restricted" within. Set the width of the restricted lane `1` and the driving lane `2` next to it to form the traffic island.

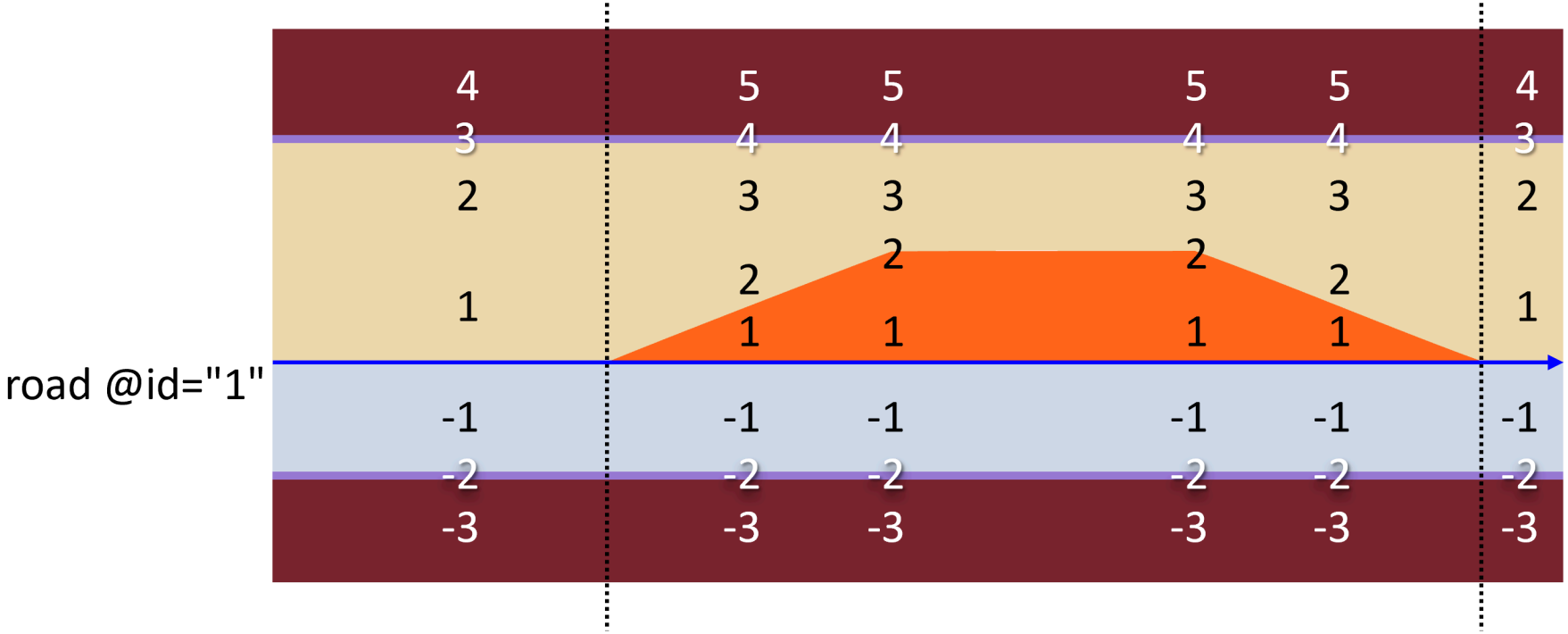


Figure 62. Restricted lane and changed lane width

Figure 62 shows the road for general traffic with the restricted part in the middle.

Code 51. Restricted lane and changed lane width

```xml
<road name="drivingRoad" length="21" id="1">
    ...
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left>
                <lane id="4" type="walking">
                    ...
                </lane>
                <lane id="3" type="curb">
                    ...
                </lane>
                <lane id="2" type="driving">
                    ...
                </lane>
                <lane id="1" type="driving">
                    ...
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    ...
                </lane>
                <lane id="-2" type="curb">
                    ...
                </lane>
                <lane id="-3" type="walking">
                    ...
                </lane>
            </right>
        </laneSection>
        <laneSection s="4.0000000000000000e+01">
            <left>
                <lane id="5" type="walking">
                    ...
                </lane>
```

```
                <lane id="4" type="curb">
                    ...
                </lane>
                <lane id="3" type="driving">
                    ...
                </lane>
                <lane id="2" type="driving">
                    <width sOffset="0.0000000000000000e+00"
                           a="3.7000000000000002e+00"
                           b="0.0000000000000000e+00"
                           c="-2.7749999999999997e-02"
                           d="9.2499999999999993e-04"/>
                    <width sOffset="2.0000000000000000e+01"
                           a="0.0000000000000000e+00"
                           b="0.0000000000000000e+00"
                           c="2.7749999999999997e-02"
                           d="-9.2499999999999993e-04"/>
                </lane>
                <lane id="1" type="restricted">
                    <width sOffset="0.0000000000000000e+00"
                           a="0.0000000000000000e+00"
                           b="0.0000000000000000e+00"
                           c="2.7749999999999997e-02"
                           d="-9.2499999999999993e-04"/>
                    <width sOffset="2.0000000000000000e+01"
                           a="3.7000000000000002e+00"
                           b="0.0000000000000000e+00"
                           c="-2.7749999999999997e-02"
                           d="9.2499999999999993e-04"/>
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    ...
                </lane>
                <lane id="-2" type="curb">
```

```
                    ...
                </lane>
                <lane id="-3" type="walking">
                    ...
                </lane>
            </right>
        </laneSection>
        <laneSection s="8.0000000000000000e+01">
            <left>
                <lane id="4" type="walking">
                    ...
                </lane>
                <lane id="3" type="curb">
                    ...
                </lane>
                <lane id="2" type="driving">
                    <width sOffset="0.0000000000000000e+00"
                           a="3.7000000000000002e+00"
                           b="0.0000000000000000e+00"
                           c="0.0000000000000000e+00"
                           d="0.0000000000000000e+00"/>
                </lane>
                <lane id="1" type="driving">
                    <width sOffset="0.0000000000000000e+00"
                           a="3.7000000000000002e+00"
                           b="0.0000000000000000e+00"
                           c="0.0000000000000000e+00"
                           d="0.0000000000000000e+00"/>
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    ...
                </lane>
                <lane id="-2" type="curb">
                    ...
```

```
                </lane>
                <lane id="-3" type="walking">
                    ...
                </lane>
            </right>
        </laneSection>
    </lanes>
    ...
</road>
```

```
7
9
8
0
8
1
8
2
8
3
8
4
8
5
8
6
8
7
8
8
8
9
9
0
9
1
9
2
9
3
9
4
9
5
9
6
9
```

```
7
9
8
9
9
1
0
0
1
0
1
1
0
2
1
0
3
1
0
4
1
0
5
1
0
6
1
0
7
1
0
8
1
0
9
1
1
```

```
0
1
1
1
1
1
2
1
1
3
1
1
4
1
1
5
1
1
6
1
1
7
1
1
8
1
1
9
```

4. Set the successors and predecessors of the lanes.

Code 52. Links between the lanes of different lane sections

```xml
<road name="drivingRoad" length="1.2000000000000000e+02" id="1">
    ...
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left>
                <lane id="4" type="walking">
                    <link>
                        <successor id="5"/>
                    </link>
                </lane>
                <lane id="3" type="curb">
                    <link>
                        <successor id="4"/>
                    </link>
                </lane>
                <lane id="2" type="driving">
                    <link>
                        <successor id="3"/>
                    </link>
                </lane>
                <lane id="1" type="driving">
                    <link>
                        <successor id="2"/>
                    </link>
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    <link>
                        <successor id="-1"/>
                    </link>
                </lane>
                <lane id="-2" type="curb">
                    <link>
                        <successor id="-2"/>
```

```
                    </link>
                </lane>
                <lane id="-3" type="walking">
                    <link>
                        <successor id="-3"/>
                    </link>
                </lane>
            </right>
        </laneSection>
        <laneSection s="4.0000000000000000e+01">
            <left>
                <lane id="5" type="walking">
                    <link>
                        <predecessor id="4"/>
                        <successor id="4"/>
                    </link>
                </lane>
                <lane id="4" type="curb">
                    <link>
                        <predecessor id="3"/>
                        <successor id="3"/>
                    </link>
                </lane>
                <lane id="3" type="driving">
                    <link>
                        <predecessor id="2"/>
                        <successor id="2"/>
                    </link>
                </lane>
                <lane id="2" type="driving">
                    <link>
                        <predecessor id="1"/>
                        <successor id="1"/>
                    </link>
                    <width ... />
                    <width ... />
                </lane>
```

```
4                   <lane id="1" type="restricted">
2                       <width ... />
4                       <width ... />
3                   </lane>
4               </left>
4               <center>...</center>
4               <right>
5                   <lane id="-1" type="driving">
4                       <link>
6                           <predecessor id="-1"/>
4                           <successor id="-1"/>
7                       </link>
4                   </lane>
8                   <lane id="-2" type="curb">
4                       <link>
9                           <predecessor id="-2"/>
5                           <successor id="-2"/>
0                       </link>
5                   </lane>
1                   <lane id="-3" type="walking">
5                       <link>
2                           <predecessor id="-3"/>
5                           <successor id="-3"/>
3                       </link>
5                   </lane>
4               </right>
5           </laneSection>
5           <laneSection s="8.0000000000000000e+01">
5               <left>
6                   <lane id="4" type="walking">
5                       <link>
7                           <predecessor id="5"/>
5                       </link>
8                   </lane>
5                   <lane id="3" type="curb">
9                       <link>
6                           <predecessor id="4"/>
```

```
                    </link>
                </lane>
                <lane id="2" type="driving">
                    <link>
                        <predecessor id="3"/>
                    </link>
                    <width ... />
                </lane>
                <lane id="1" type="driving">
                    <link>
                        <predecessor id="2"/>
                    </link>
                    <width ... />
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    <link>
                        <predecessor id="-1"/>
                    </link>
                </lane>
                <lane id="-2" type="curb">
                    <link>
                        <predecessor id="-2"/>
                    </link>
                </lane>
                <lane id="-3" type="walking">
                    <link>
                        <predecessor id="-3"/>
                    </link>
                </lane>
            </right>
        </laneSection>
    </lanes>
    ...
</road>
```

```
7
9
8
0
8
1
8
2
8
3
8
4
8
5
8
6
8
7
8
8
8
9
9
0
9
1
9
2
9
3
9
4
9
5
9
6
9
```

```
7
9
8
9
9
1
0
0
1
0
1
1
0
2
1
0
3
1
0
4
1
0
5
1
0
6
1
0
7
1
0
8
1
0
9
1
1
```

```
0
1
1
1
1
1
2
1
1
3
1
1
4
1
1
5
1
1
6
1
1
7
1
1
8
1
1
9
1
2
0
1
2
1
1
2
2
```

```
1
2
3
1
2
4
1
2
5
1
2
6
1
2
7
1
2
8
1
2
9
1
3
0
1
3
1
1
3
2
1
3
3
1
3
4
1
```

```
3
5
1
3
6
1
3
7
1
3
8
1
3
9
1
4
0
1
4
1
1
4
2
1
4
3
1
4
4
1
4
5
1
4
6
1
```

```
4
7
```

5. Place the road reference line for the road for the pedestrian crossing. Start and end the road reference line in the middle of the lanes with @type="walking" that will be connected by the pedestrian crossing.

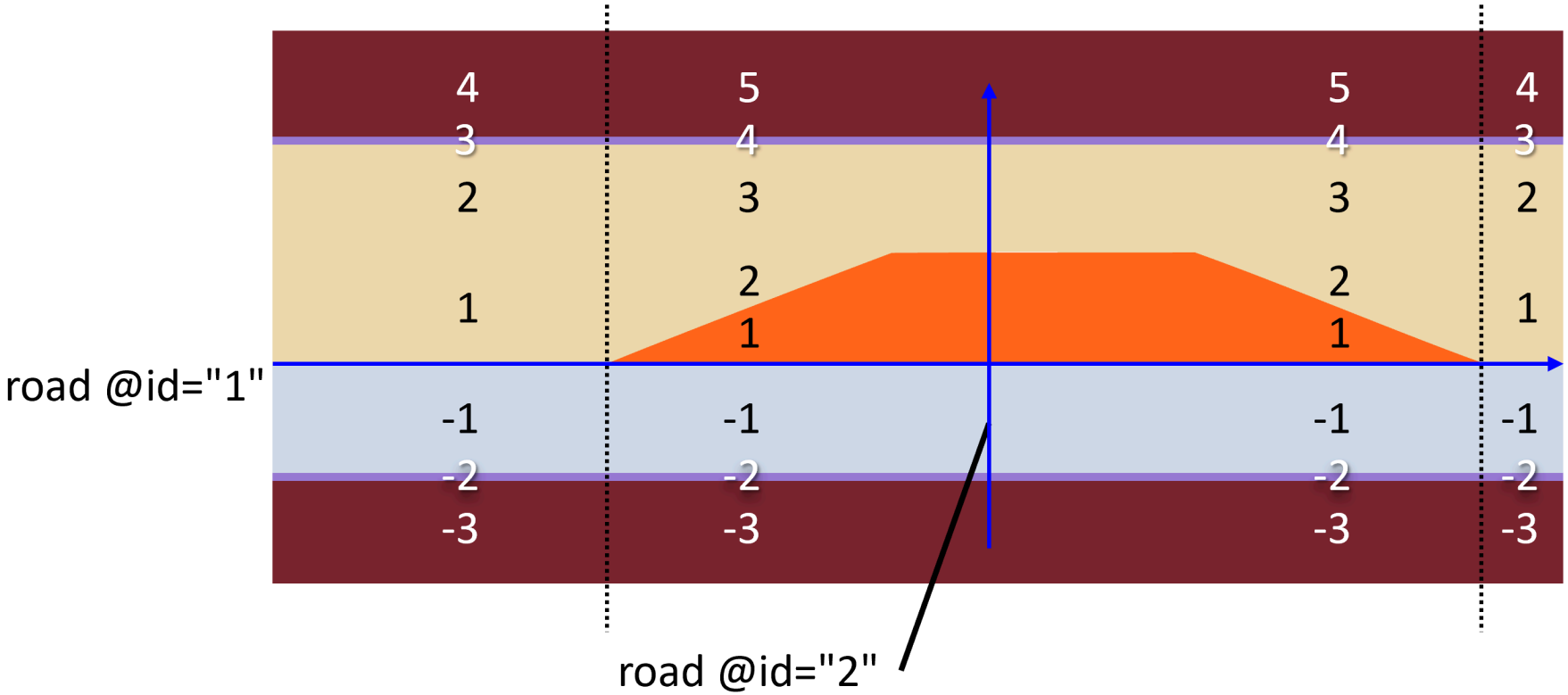


Figure 63. Road reference line of the pedestrian crossing

Figure 63 shows the road reference line of the road for the pedestrian crossing.

Code 53. Road for the pedestrian crossing

```
<road name="pedestrian" length="1.4000000000000000e+01" id="2">
    ...
</road>
```

6. Add one lane with @type="walking" on the road for the pedestrian crossing.

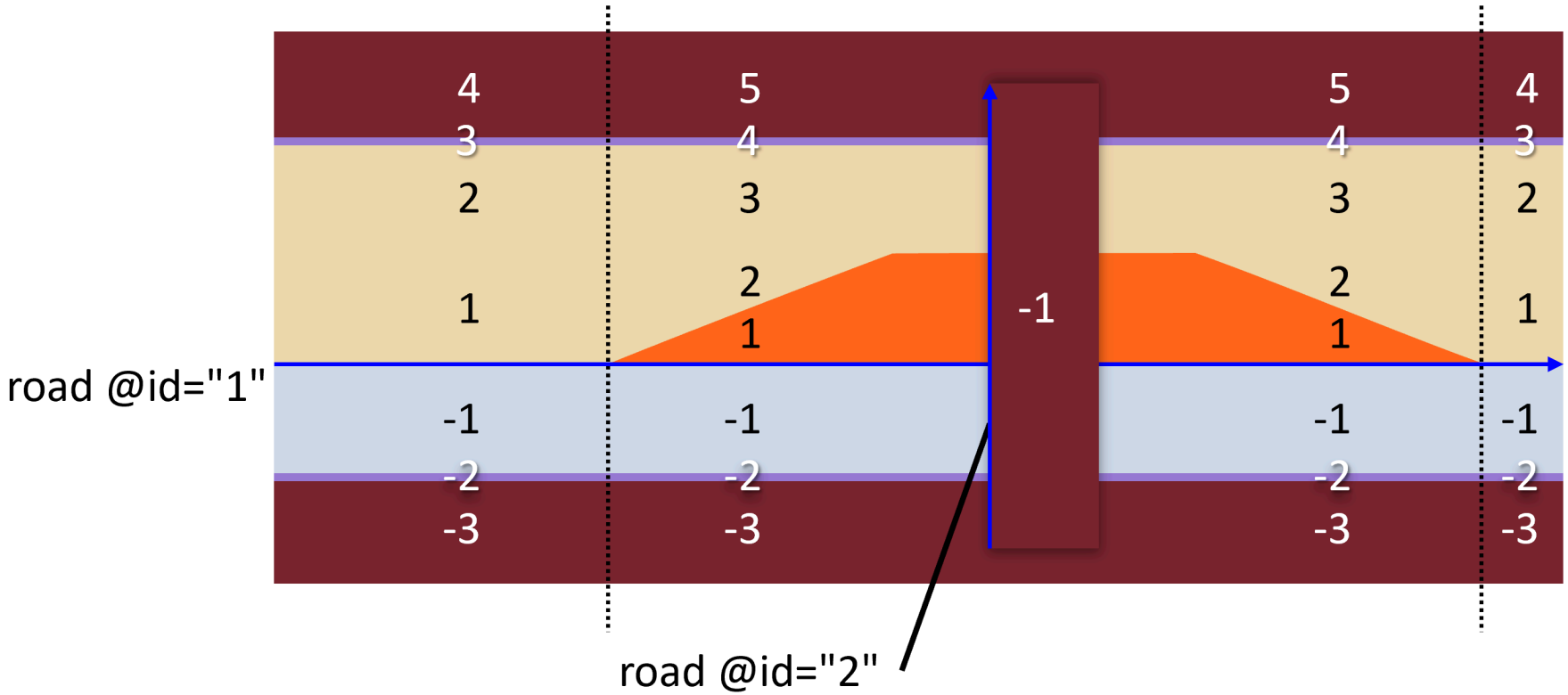


Figure 64. All lanes of the pedestrian crossing

Figure 64 shows the roads of the pedestrian crossing with all applied lane types.

Code 54. All roads and lanes of the pedestrian crossing

```xml
<road name="drivingRoad" length="1.2000000000000000e+02" id="1">
    ...
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left>
                <lane id="4" type="walking">
                    ...
                </lane>
                <lane id="3" type="curb">
                    ...
                </lane>
                <lane id="2" type="driving">
                    ...
                </lane>
                <lane id="1" type="driving">
                    ...
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    ...
                </lane>
                <lane id="-2" type="curb">
                    ...
                </lane>
                <lane id="-3" type="walking">
                    ...
                </lane>
            </right>
        </laneSection>
        <laneSection s="4.0000000000000000e+01">
            <left>
                <lane id="5" type="walking">
                    <link>
                        <predecessor id="4"/>
```

```
                        <successor id="4"/>
                    </link>
                </lane>
                <lane id="4" type="curb">
                    <link>
                        <predecessor id="3"/>
                        <successor id="3"/>
                    </link>
                </lane>
                <lane id="3" type="driving">
                    <link>
                        <predecessor id="2"/>
                        <successor id="2"/>
                    </link>
                </lane>
                <lane id="2" type="driving">
                    <link>
                        <predecessor id="1"/>
                        <successor id="1"/>
                    </link>
                    <width ... />
                    <width ... />
                </lane>
                <lane id="1" type="restricted">
                    <width ... />
                    <width ... />
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    <link>
                        <predecessor id="-1"/>
                        <successor id="-1"/>
                    </link>
                </lane>
                <lane id="-2" type="curb">
```

```
4                       <link>
2                           <predecessor id="-2"/>
4                           <successor id="-2"/>
3                       </link>
4                   </lane>
4                   <lane id="-3" type="walking">
4                       <link>
5                           <predecessor id="-3"/>
4                           <successor id="-3"/>
6                       </link>
4                   </lane>
7               </right>
4           </laneSection>
8           <laneSection s="8.0000000000000000e+01">
4               <left>
9                   <lane id="4" type="walking">
5                       <link>
0                           <predecessor id="5"/>
5                       </link>
1                   </lane>
5                   <lane id="3" type="curb">
2                       <link>
5                           <predecessor id="4"/>
3                       </link>
5                   </lane>
4                   <lane id="2" type="driving">
5                       <link>
5                           <predecessor id="3"/>
5                       </link>
6                       <width ... />
5                   </lane>
7                   <lane id="1" type="driving">
5                       <link>
8                           <predecessor id="2"/>
5                       </link>
9                       <width ... />
6                   </lane>
```

```
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    <link>
                        <predecessor id="-1"/>
                    </link>
                </lane>
                <lane id="-2" type="curb">
                    <link>
                        <predecessor id="-2"/>
                    </link>
                </lane>
                <lane id="-3" type="walking">
                    <link>
                        <predecessor id="-3"/>
                    </link>
                </lane>
            </right>
        </laneSection>
    </lanes>
    ...
</road>
<road name="pedestrian" length="1.4000000000000000e+01" id="2">
    ...
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left/>
            <center>...</center>
            <right>
                <lane id="-1" type="walking">
                    ...
                </lane>
            </right>
        </laneSection>
    </lanes>

```

```
    ...
</road>
```

```
7
9
8
9
9
1
0
0
1
0
1
1
0
2
1
0
3
1
0
4
1
0
5
1
0
6
1
0
7
1
0
8
1
0
9
1
1
```

```
0
1
1
1
1
1
2
1
1
3
1
1
4
1
1
5
1
1
6
1
1
7
1
1
8
1
1
9
1
2
0
1
2
1
1
2
2
```

```
1
2
3
1
2
4
1
2
5
1
2
6
1
2
7
1
2
8
1
2
9
1
3
0
1
3
1
1
3
2
1
3
3
1
3
4
1
```

```
3
5
1
3
6
1
3
7
1
3
8
1
3
9
1
4
0
1
4
1
1
4
2
1
4
3
1
4
4
1
4
5
1
4
6
1
4
```

```
7
1
4
8
```

7. Create the junction with @type="virtual". Set the value of the @mainRoad attribute to the id of the road for general traffic. Set the values of the @sStart and @sEnd attributes to be clearly outside of the area of crossing traffic.

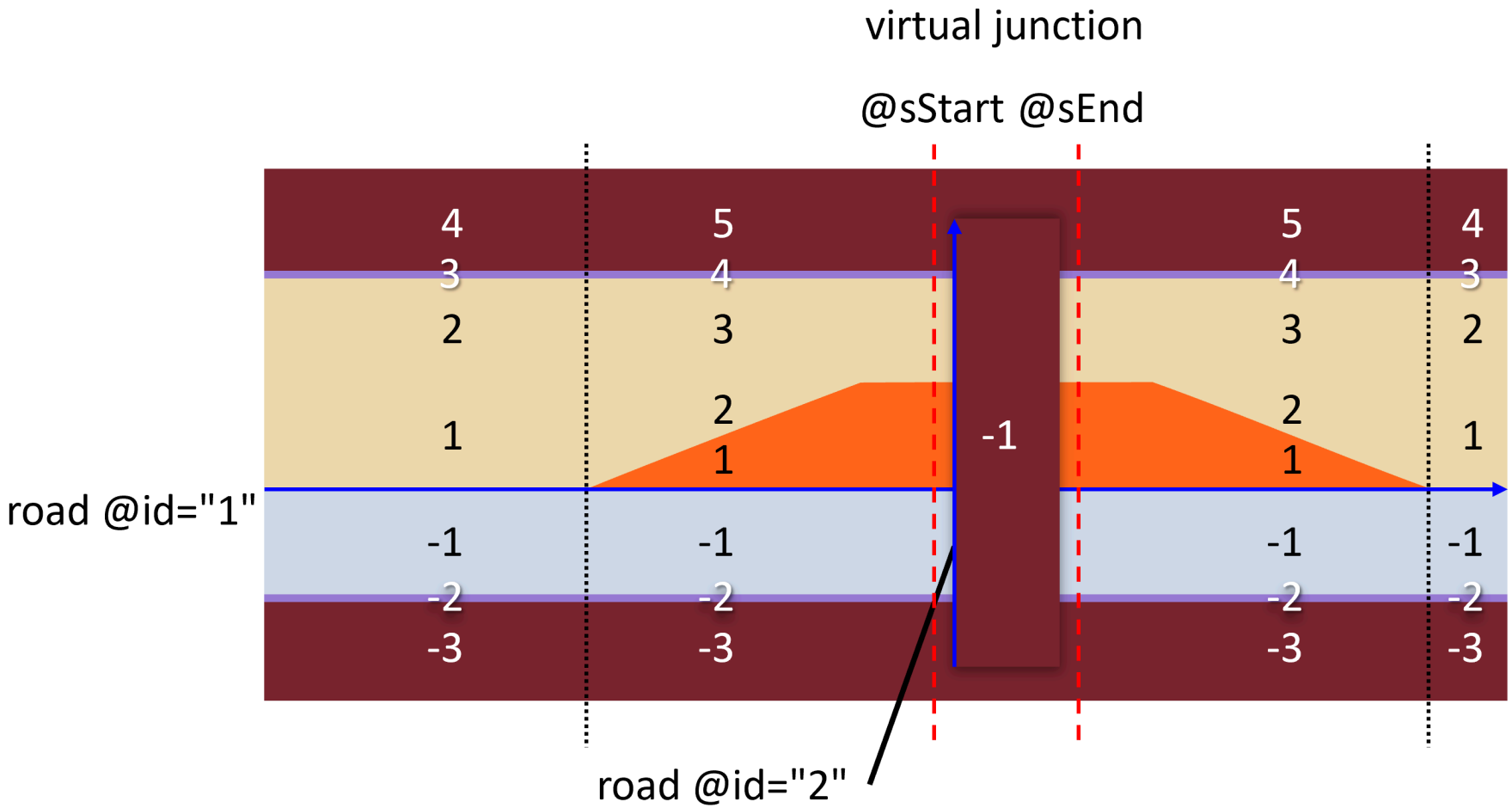


Figure 65. Virtual junction for the pedestrian crossing

Figure 65 shows the virtual junction around the area of crossing traffic.

Code 55. Junction with @type="virtual"

```
<junction name="pedestrianCrossPath" type="virtual" id="555" mainRoad="1" sStart="5.7500000000000000e+01" sEnd=
    ...
</junction>
```

8. Create a cross path in the junction. Set the value of the @crossingRoad attribute to the id of the road crossing the main road of the junction. Set the values of the @roadAtStart and @roadAtEnd attributes to the id of the roads connected by the cross path.

Code 56. Cross path in the junction

```
<junction name="pedestrianCrossPath" type="virtual" id="555" mainRoad="1" sStart="5.7500000000000000e+01" sEnd=
    <crossPath id="0" crossingRoad="2" roadAtStart="1" roadAtEnd="1">
        ...
    </crossPath>
</junction>
```

9. Create lane links in the cross path to connect the lanes. Set the value of the @from attribute to the id of the lane on the road at start or end of the cross path. Set the value of the @to attribute to the id of the lane at the crossing road.

Code 57. Cross path with lane links

```
<junction name="pedestrianCrossPath" type="virtual" id="555" mainRoad="1" sStart="5.7500000000000000e+01" sEnd=
    <crossPath id="0" crossingRoad="2" roadAtStart="1" roadAtEnd="1">
        <startLaneLink s="5.9000000000000000e+01" from="5" to="-1"/>
        <endLaneLink s="5.9000000000000000e+01" from="-3" to="-1"/>
    </crossPath>
</junction>
```

10. Set the priority. Set the value of the @high attribute to the id of the road for general traffic and set the value of the @low attribute to the id of the road for the pedestrian crossing.

Code 58. Junction with priority

```
<junction name="pedestrianCrossPath" type="virtual" id="555" mainRoad="1" sStart="5.7500000000000000e+01" sEnd=
    <priority high="1" low="2"/>
    <crossPath id="0" crossingRoad="2" roadAtStart="1" roadAtEnd="1">
        <startLaneLink s="5.9000000000000000e+01" from="5" to="-1"/>
        <endLaneLink s="5.9000000000000000e+01" from="-3" to="-1"/>
    </crossPath>
</junction>
```

11. Set the value of the @junction attribute at the road for the pedestrian crossing to the id of the created virtual junction.

Code 59. Roads at the pedestrian crossing with @junction attributes

```
<road name="drivingRoad" length="21" id="1" junction="-1">
    ...
</road>
<road name="pedestrian" length="12" id="2" junction="555">
    ...
</road>
```

The @junction attribute at the road for general traffic has the value `-1` because the road is not part of the junction.

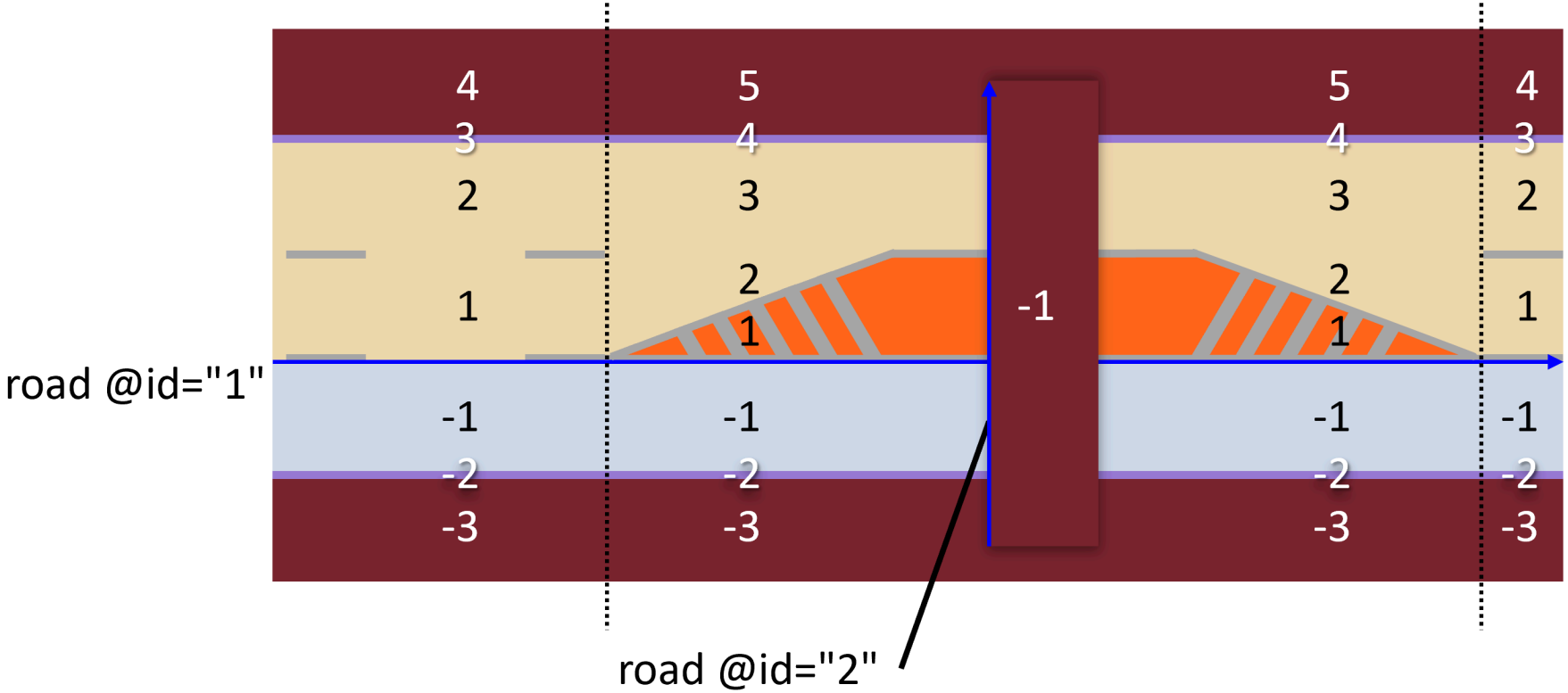


Figure 66. Complete pedestrian crossing

Figure 66 shows a complete pedestrian crossing.

Code 60. Complete pedestrian crossing

```xml
<road name="drivingRoad" length="1.2000000000000000e+02" id="1" junction="-1">
    <link> ...</link>
    <planView>
        <geometry>
            ...
        </geometry>
    </planView>
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left>
                <lane id="4" type="walking">
                    <link>...</link>
                </lane>
                <lane id="3" type="curb">
                    <link>...</link>
                </lane>
                <lane id="2" type="driving">
                    <link>...</link>
                </lane>
                <lane id="1" type="driving">
                    <link>...</link>
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving">
                    <link>...</link>
                </lane>
                <lane id="-2" type="curb">
                    <link>...</link>
                </lane>
                <lane id="-3" type="walking">
                    <link>...</link>
                </lane>
            </right>
        </laneSection>
```

```
        <laneSection s="4.0000000000000000e+01">
            <left>
                <lane id="4" type="walking">
                    <link>...</link>
                </lane>
                <lane id="3" type="curb">
                    <link>...</link>
                </lane>
                <lane id="2" type="driving">
                    <link>...</link>
                </lane>
                <lane id="1" type="restricted">
                    <link>...</link>
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving" level="false">
                    <link>...</link>
                </lane>
                <lane id="-2" type="curb">
                    <link>...</link>
                </lane>
                <lane id="-3" type="walking">
                    <link>...</link>
                </lane>
            </right>
        </laneSection>
        <laneSection s="8.0000000000000000e+01">
            <left>
                <lane id="4" type="walking">
                    <link>...</link>
                </lane>
                <lane id="3" type="curb">
                    <link>...</link>
                </lane>
                <lane id="2" type="driving">
```

```
                    <link>...</link>
                </lane>
                <lane id="1" type="driving">
                    <link>...</link>
                </lane>
            </left>
            <center>...</center>
            <right>
                <lane id="-1" type="driving" level="false">
                    <link>...</link>
                </lane>
                <lane id="-2" type="curb">
                    <link>...</link>
                </lane>
                <lane id="-3" type="walking">
                    <link>...</link>
                </lane>
            </right>
        </laneSection>
    </lanes>
    ...
</road>
<road name="pedestrian" length="1.4000000000000000e+01" id="2" junction="555">
    <link>...</link>
    <lanes>
        <laneSection s="0.0000000000000000e+00">
            <left/>
            <center>...</center>
            <right>
                <lane id="-1" type="walking">
                    <link/>
                </lane>
            </right>
        </laneSection>
    </lanes>
</road>
...
```

```
<junction name="pedestrianCrossPath" type="virtual" id="555" mainRoad="1" sStart="5.7500000000000000e+01" sEnd=
    <priority high="1" low="2"/>
    <crossPath id="0" crossingRoad="2" roadAtStart="1" roadAtEnd="1">
        <startLaneLink s="5.4000000000000000e+01" from="4" to="-1"/>
        <endLaneLink s="5.4000000000000000e+01" from="-3" to="-1"/>
    </crossPath>
</junction>
```

```
7
9
8
0
8
1
8
2
8
3
8
4
8
5
8
6
8
7
8
8
8
9
9
0
9
1
9
2
9
3
9
4
9
5
9
6
9
```

```
7
9
8
9
9
1
0
0
1
0
1
1
0
2
1
0
3
1
0
4
1
0
5
1
0
6
1
0
7
1
0
8
1
0
9
1
1
```

```
0
1
1
1
1
1
2
1
1
3
1
1
4
1
1
5
1
1
6
1
1
7
```

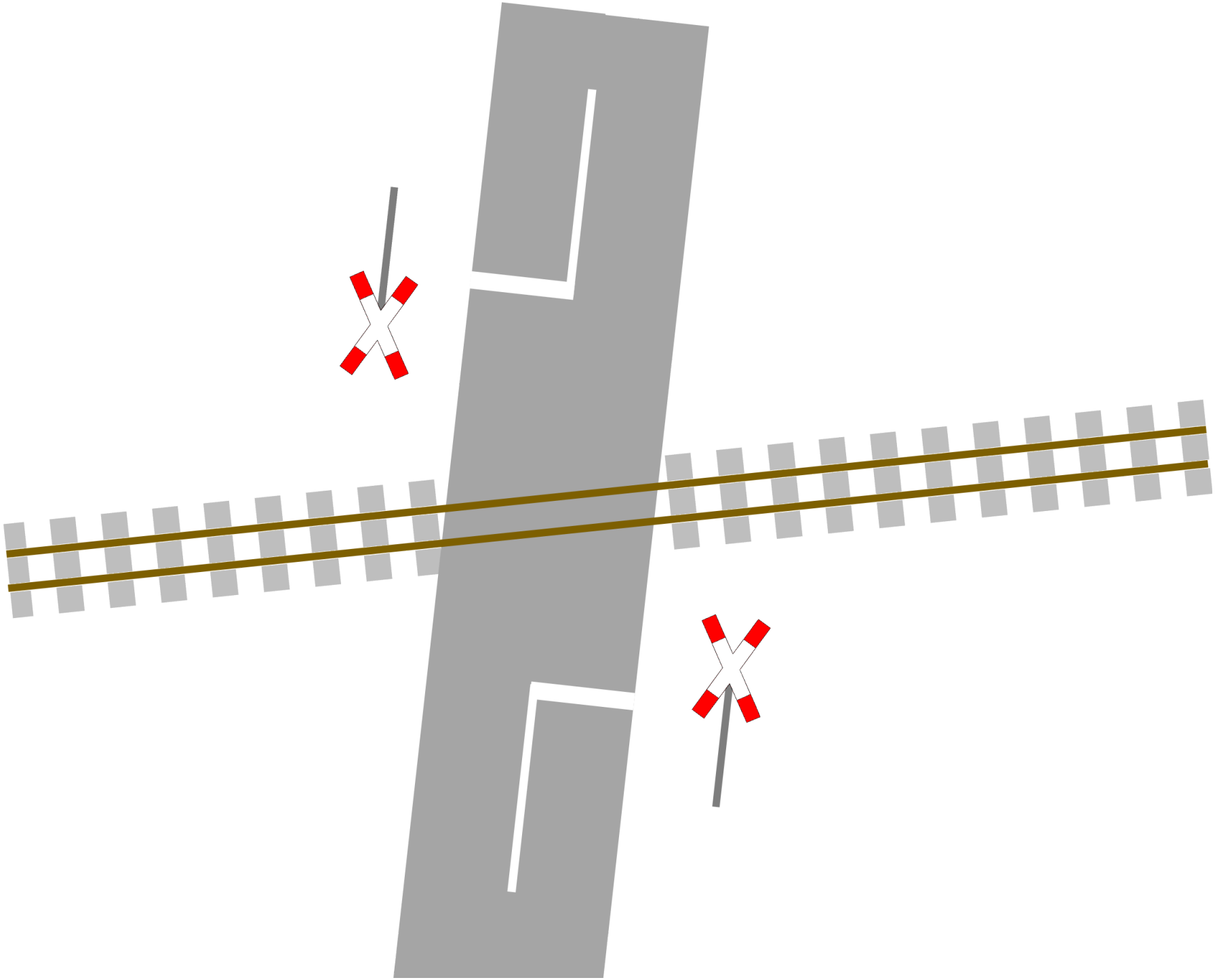
XML example

Solution in ASAM OpenDRIVE 1.8.0

- Ex_Pedestrian_Crossing.xodr

10.4 Rail road crossing

10.4.1 Example



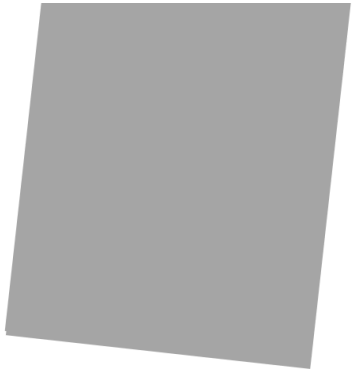


Figure 67. A rail road crossing

Figure 67 shows a rail road crossing where a railroad track with priority crosses a road for normal traffic.

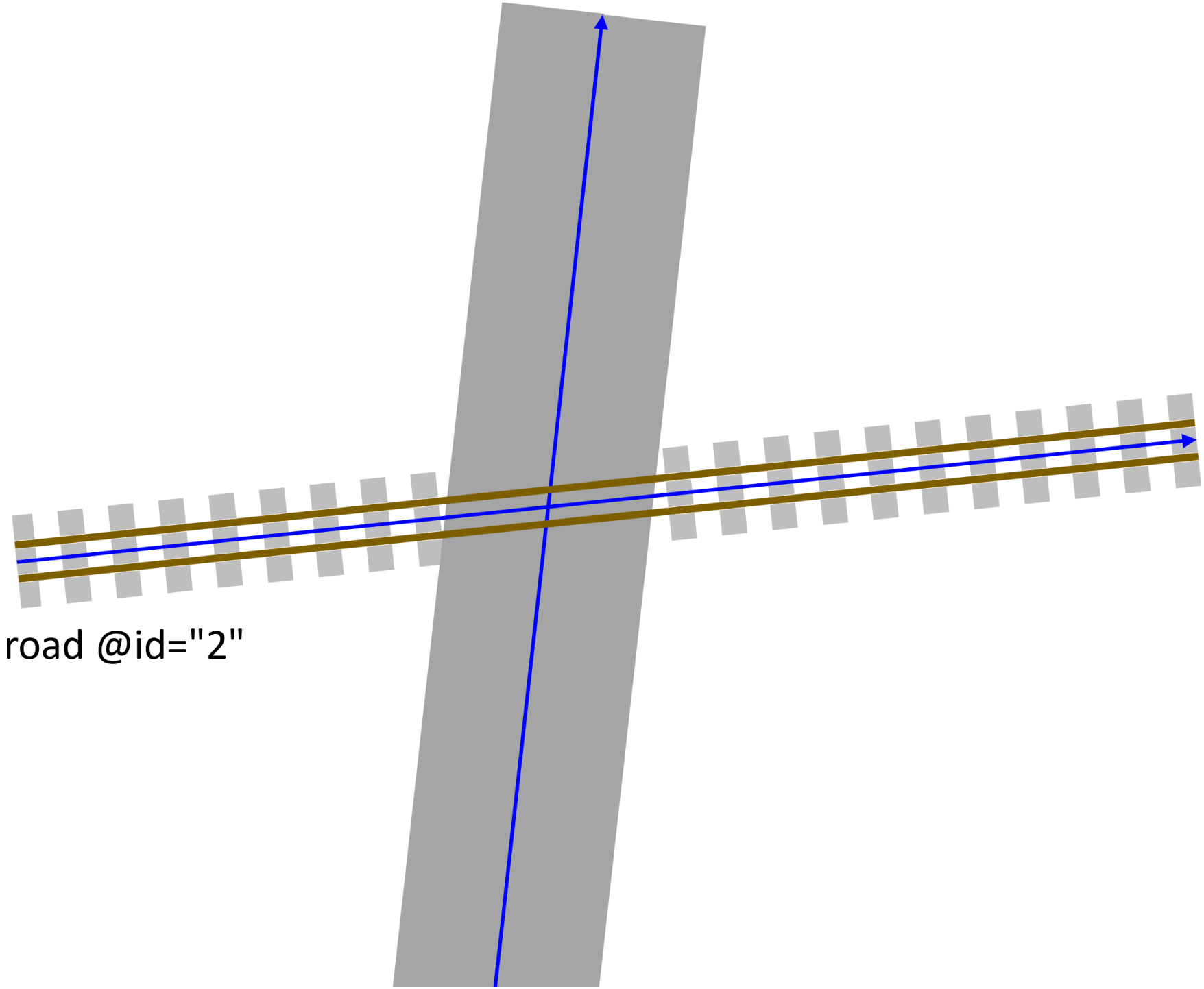
10.4.2 Challenges

- Choose between a crossing or a cross path to find the optimal junction type.
- Assign the correct lane types.
- Set the range for the overlapping lanes.
- Set the priority.
- Add road marks.

10.4.3 Modeling approach

A junction with @type="crossing" is the best choice if traffic cannot change the roads.

1. Place the road reference lines for the road that becomes the rail road track and the road that is for normal vehicle traffic.


road @id="2"

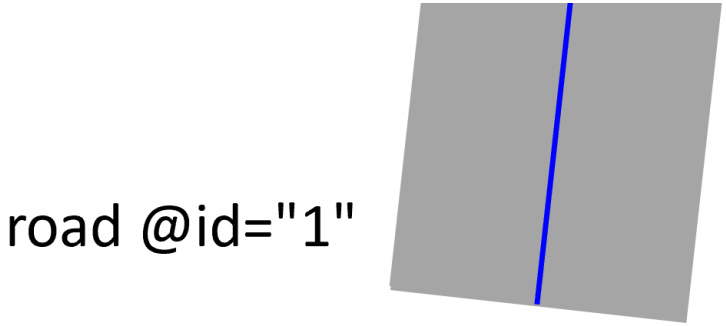


Figure 68. Road reference lines

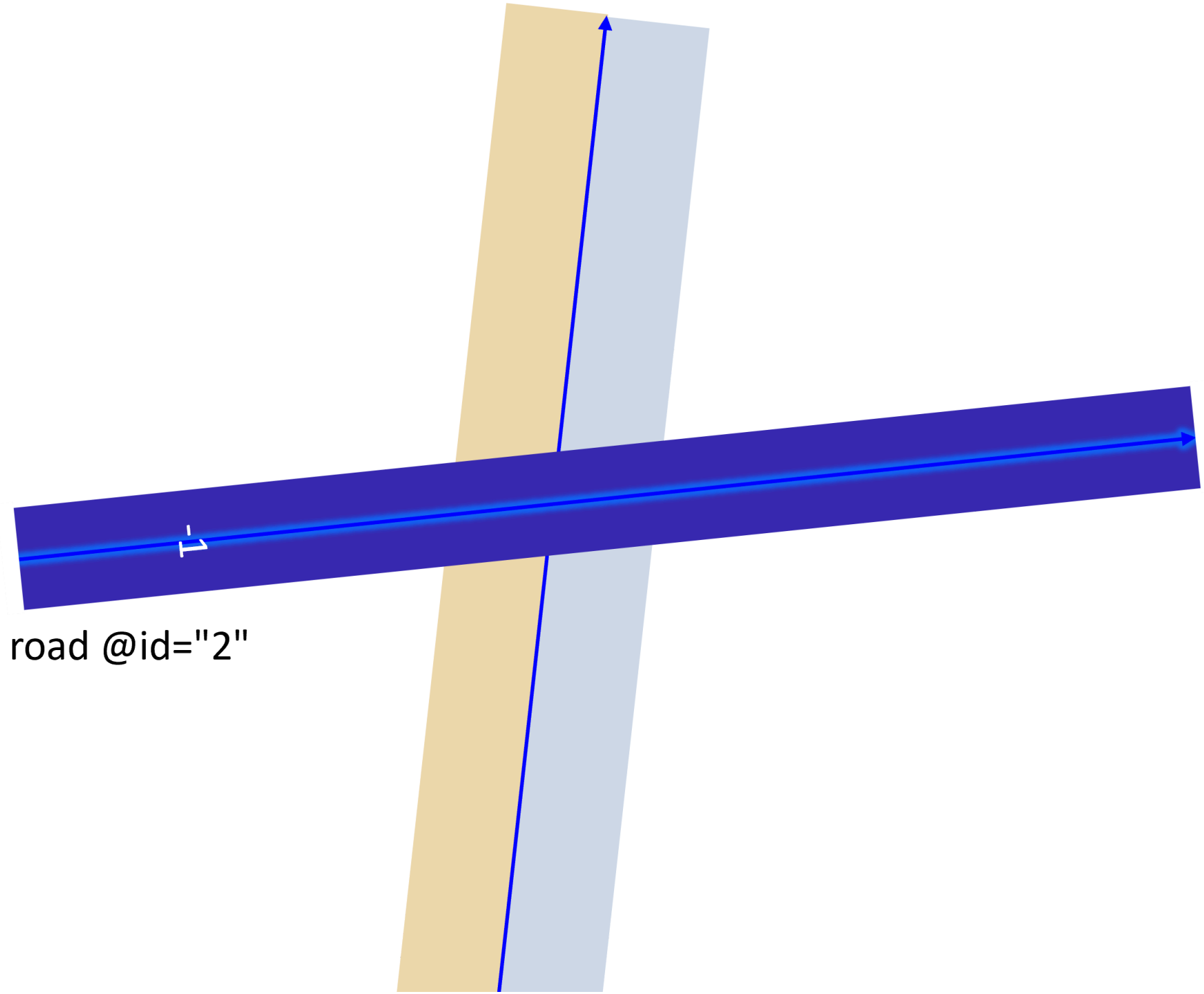
Figure 68 shows the road reference lines (blue arrows).

Code 61. Roads meeting at the crossing

```
<road name="" length="200" id="1" junction="-1">
    ...
</road>
<road name="" length="2300" id="2" junction="-1">
    ...
</road>
```

The @junction attribute at the roads has the value `-1` because the roads are not part of a junction.

2. Add all lanes for the roads and set the lane types. Add one lane with @type="driving" for each direction on the road for normal vehicle traffic. Add one lane with @type="rail" on the road for rail-bound traffic.


1
road @id="2"

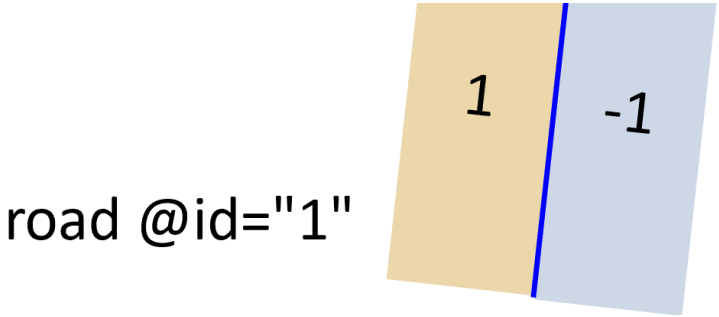


Figure 69. Lanes on the roads

Figure 69 shows the roads of the crossing with all applied lane types.

Code 62. Lanes on the roads

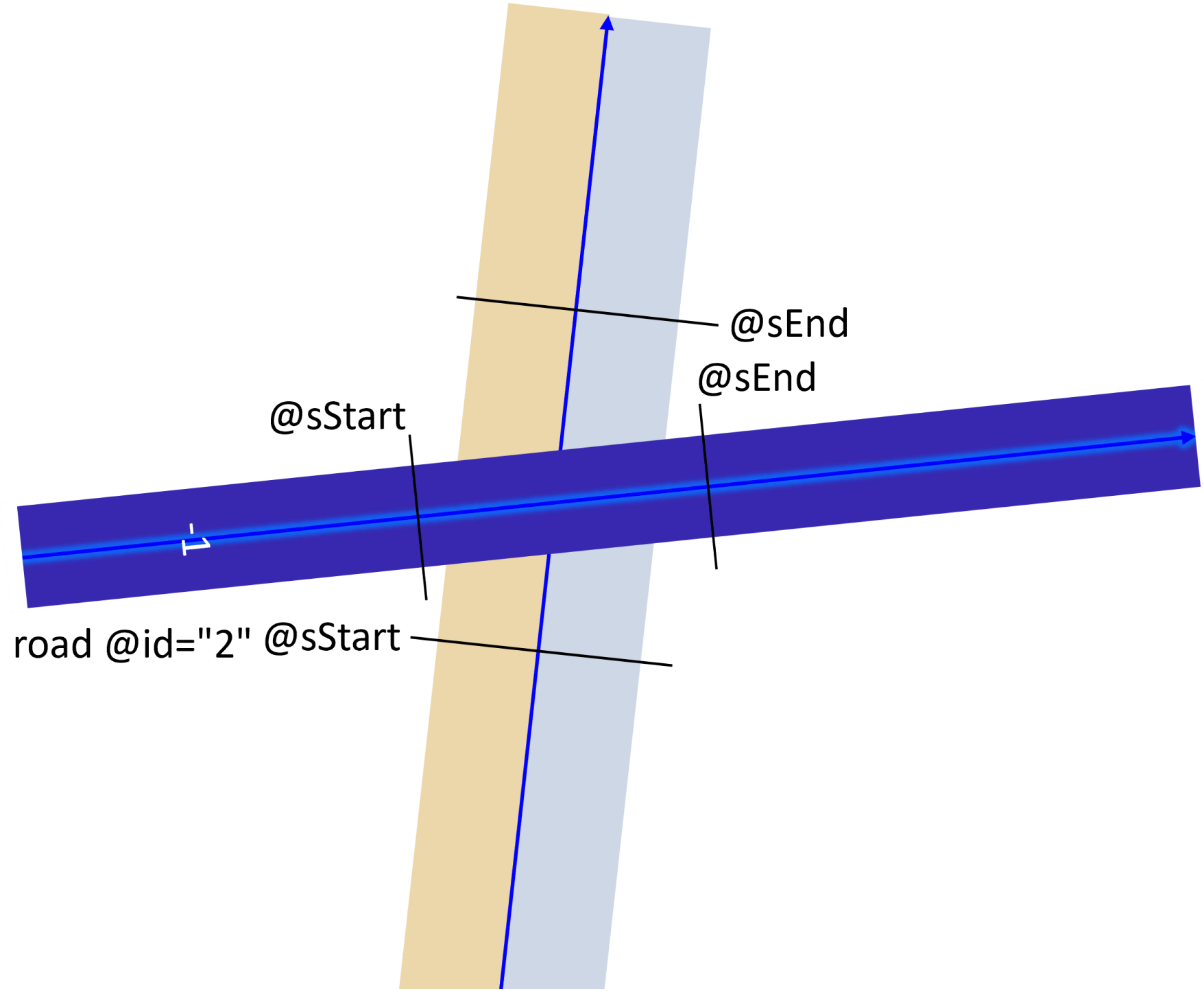
```
<road name="" length="200" id="1" junction="-1">
    <laneSection s="0.0000000000000000e+00">
        <left>
            <lane id="1" type="driving"/>
        </left>
        <center/>
        <right>
            <lane id="-1" type="driving"/>
        </right>
    </laneSection>
</road>
<road name="" length="2300" id="2" junction="-1">
    <laneSection s="0.0000000000000000e+00">
        <left/>
        <center/>
        <right>
            <lane id="-1" type="rail"/>
        </right>
    </laneSection>
</road>
```

3. Create the junction with @type="crossing".

Code 63. Junction with @type="crossing"

```
<junction name="myRailCrossing" type="crossing" id="555">
    ...
</junction>
```

4. Create one road section for each road of the crossing and set the values of the @sStart and @sEnd attributes to be clearly outside of the area of crossing traffic.


@sEnd
@sEnd
@sStart
1
road @id="2" @sStart

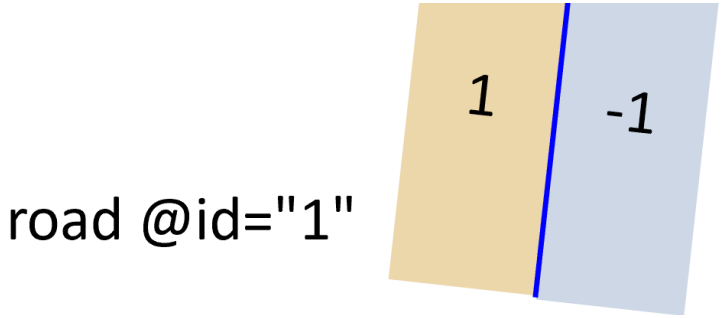


Figure 70. Road sections

Figure 70 shows the road sections and the area of crossing traffic.

Code 64. Road sections

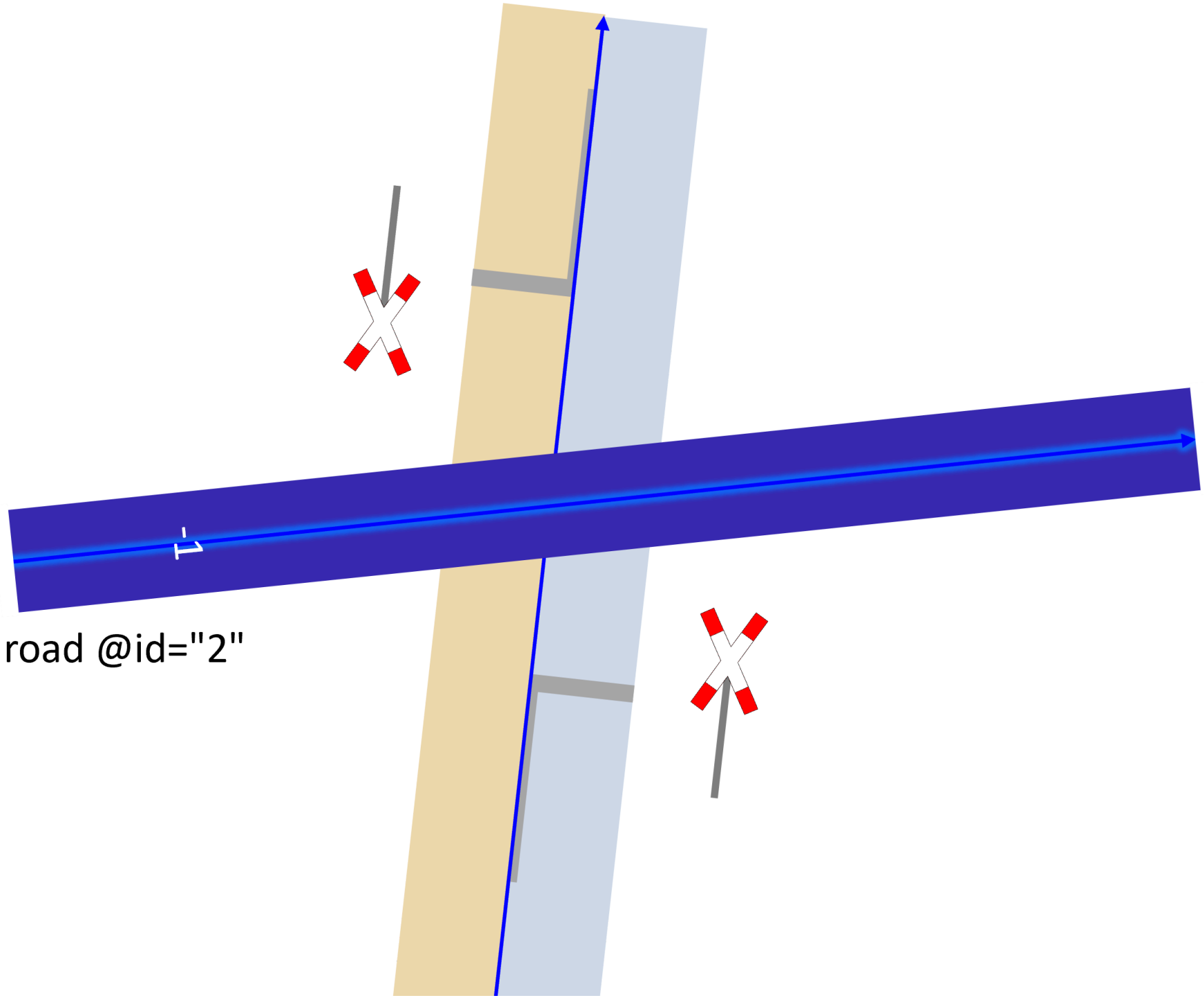
```
<junction name="myRailCrossing" type="crossing" id="555">
    <roadSection id="0" roadId="1" sStart="50" sEnd="60"/>
    <roadSection id="1" roadId="2" sStart="150" sEnd="160"/>
</junction>
```

5. Set the priority.

Code 65. Junction with priority

```
<junction name="myRailCrossing" type="crossing" id="555">
    <priority high="2" low="1"/>
    <roadSection id="0" roadId="1" sStart="50" sEnd="60"/>
    <roadSection id="1" roadId="2" sStart="150" sEnd="160"/>
</junction>
```

6. Add road markings and traffic signs.


1
road @id="2"

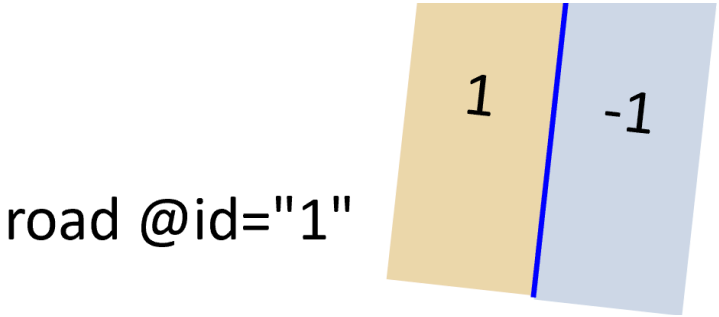


Figure 71. Complete rail road crossing

Figure 71 shows a complete rail road crossing.

Code 66. Rail road crossing

```
<road name="DrivingRoad1" length="200" id="1" junction="-1">
    <laneSection s="0.0000000000000000e+00">
        <left>
            <lane id="1" type="driving"/>
        </left>
        <center/>
        <right>
            <lane id="-1" type="driving"/>
        </right>
    </laneSection>
</road>
<road name="RAILRoad" length="2300" id="2" junction="-1">
    <laneSection s="0.0000000000000000e+00">
        <left/>
        <center/>
        <right>
            <lane id="-1" type="rail"/>
        </right>
    </laneSection>
</road>
...
<junction name="myRailCrossing" type="crossing" id="555">
    <priority high="2" low="1"/>
    <roadSection id="0" roadId="1" sStart="50" sEnd="60"/>
    <roadSection id="1" roadId="2" sStart="150" sEnd="160"/>
</junction>
```

```
3
2
4
2
5
2
6
```

10.5 Related topics

- ASAM OpenDRIVE Specification, Cross paths
- ASAM OpenDRIVE Specification, Cross paths with virtual junctions
- ASAM OpenDRIVE Specification, Crossings